

# AI-based predictive data management

Artificial Intelligence (AI)-based predictive data management is an innovative approach that uses advanced algorithms, particularly generative models, to forecast future data storage needs and enhance data management practices. While the primary focus is on data storage optimization, this technology also plays a crucial role in improving data quality and organizational efficiency. In this reading, you'll learn about the predictive capabilities of generative models in data storage and their broader applications in data quality enhancement and cataloging.

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## Generative models in data forecasting

Generative models are a type of AI that focuses on generating new data points or predictions based on existing data. Unlike machine learning models that rely on historical data to make predictions, generative models can simulate possible future scenarios by using probability techniques and pattern recognition to extrapolate potential outcomes. This ability allows businesses to prepare for various potential outcomes, including data storage and management challenges.

## Predictive analytics versus generative AI

Predictive analytics is usually based on traditional machine learning and it focuses on analyzing historical data to make future predictions. For example, cloud data analysts might use predictive analytics to forecast server load based on past usage patterns, helping in resource allocation and preventing server overloads. Generative AI (GenAI) goes beyond this to simulate potential future events to offer a broader range of outcomes and scenarios. For example, in unusual circumstances like unexpected spikes in web traffic during promotional events, analysts could predict server load usage patterns and create contingency plans. This difference is important for understanding the capabilities and limitations of both technologies.

## Enhancing data quality and cataloging

Beyond forecasting future data storage needs, GenAI can improve data quality. It can fill in missing data points or augment existing data, improving the overall reliability of the data set. This capability is useful for when data is incomplete or sparse.

## Gemini for Workspace with BigQuery

Imagine you're a cloud data analyst that needs to create queries to identify recent anomalies in a data set. You can utilize BigQuery's integrated tool, Gemini for Workspace to accomplish this task. Together, BigQuery and Gemini for Workspace can help you more efficiently interact with data, write enhanced queries, and receive comprehensive SQL query result explanations. This

type of AI use can help produce more accurate and faster data retrieval, crucial to help you identify underlying patterns and anomalies for timely decision-making. You can also use Gemini for Workspace's capability of query assistance and explanation to further explore data, making it a helpful tool in predictive data management and enhancing the overall efficiency and effectiveness of your data analysis process.

## Create sample data for testing

Now let's suppose you've been tasked to create sample data to make predictions for a product still in development, where there's no actual user data yet. AI can be useful for creating sample data for testing purposes. GenAI can generate realistic and diverse datasets that closely mimic real data. This capability is especially useful where access to actual data is limited due to privacy and ethical concerns or availability issues. For example, you can use GenAI to produce synthetic data that mimics the complexity and variability of genuine data, enabling you to rigorously test and validate your analytical models and tools in a controlled environment. With GenAI, you can explore methods for resolving issues before applying them to real data sets.

## Cataloging and organizational efficiency

Efficient data cataloging, facilitated by AI, ensures that data is stored properly and also easily retrievable. AI-driven cataloging systems help in organizing and classifying data more effectively, employing advanced tagging and indexing algorithms. This includes automatically classifying data based on content, usage patterns, and access frequency, optimizing storage and retrieval processes. For example, maybe you work as a cloud data analyst for a multinational corporation. You could use AI to help you categorize thousands of client contracts based on region, contract value, and renewal dates, streamlining access for legal and sales teams.

## Preparing for future data challenges with AI

GenAI helps equip businesses to respond to current data management needs and prepare for future challenges. For example, you could use AI to help anticipate the impact of new data privacy regulations, adjusting data handling and storage protocols proactively. This approach is essential in the rapidly evolving digital landscape where data volumes and complexity are constantly increasing.

## Key takeaways

GenAI models offer advanced forecasting capabilities, enabling businesses to help prepare for various data storage scenarios. Unlike traditional machine learning, GenAI can simulate possible future outcomes, providing a broader perspective. AI can enhance data quality and efficiency through improved cataloging and data management practices. Understanding the distinction between predictive analytics and GenAI is crucial in leveraging their full potential in data management.